# A Multi-Level Reputation-Based Pollution Attacks Detection and Prevention in P2P Streaming

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***Abstract*— Today Peer-To-Peer (P2P) networks are becoming more important from high scalability and low cost point of view, and these networks are popular platform for video streaming. With burgeoning of P2P networks, one of the interesting areas of research is confronting to various attacks and threats against these networks. Since, video streaming is a popular application of P2P networks, some attacks like pollution attacks have made serious challenge. In these attacks fake contents are replaced with the original one, by malicious users, and are broadcasted to the network. Diffusion of these contents cannot be controlled with any control process and causes that the polluted content quickly becomes available in network. Another major impacts of this event are users dissatisfaction performance reduction and poor quality of services. For reason, this article proposes a solution for addressing the pollution attacks on P2P networks. Our approach is based on CDN-P2P architecture which is called "Multi-Level mechanism". This mechanism, unlike produced methods, uses temporary blocked policy and new technique to calculate reputation. Also Multi-Level mechanism works in a distributed reputation system. Experimental results show that our Multi-Level approach has a good performance against pollution attacks.**

***Keywords*— Pollution Attacks, Video Streaming, Peer-to-Peer, Reputation**

## I. INTRODUCTION

In recent years we have seen a significant growth in the number of smartphones, tablets, smart TVs and so we have high-speed internet connections so that everything can be provided on the users screen. Also Internet applications have been developed rapidly without restricting the flow time sensitive multimedia. With increasing advances in Internet technologies and communication toward time sensitive multimedia applications, entertainment and scientific environments in world, the demand is taken to use them. One of the time sensitive applications is video streaming that is classified into two categories: live video and video on demand (VoD).

During the live video streaming, the users view the video simultaneously and in this type of streaming, the player time is very important. But in VoD streaming, video playback time is different between users and video frames do not require simultaneously to be available to all users. In fact, this ability causes more flexibility in playback VoD. Usually in P2P networks, clients are organized in an overlay network. For content delivery in overlay networks, there are three main approaches: pull-based, push-based and hybrid pull-push. Also overlay structures can be broadly divided into three categories: tree-based, mesh-based and hybrid tree-mesh [14] [15] [16].

One of the most serious attacks on P2P networks are pollution attacks on the video streaming that done by malicious peers. Many organizations use P2P network content distribution and complain for the resources were wasted by the attackers. Pollution attacks are formed due to malicious peers in the network and the purpose of pollution attacks is to create fake content and send it to neighbours (or other peers in the network) for decreasing system performance and disruption of the network. So if the control over these attacks (the polluted content and polluter peers) is not done within a short time, the polluted content is distributed on the network. The authors in [1] shows that if a polluter peer is existed in the network with 3300 peers and there is no control on polluter, then by distribution of infected and forged content in the network, it took 34 minutes to completely down the network. So, a security mechanism is a basic requirement that fights with pollution attacks.

In this paper, we have proposed a Multi-Level mechanism to detect and prevent polluter in pull-based and mesh-based P2P video streaming systems. The Multi-Level mechanism is easily usable in live and VoD streaming, but in this article, we implement Multi-Level on VoD streaming.

## II. FIGHTING POLLUTION ATTACKS IN P2P STREAMING

In general the fight with pollution attacks is done in three steps: detection polluted content, detection polluter peers and dealing with polluter peers.

Many methods have been proposed to fight with pollution attacks for the step detection polluted content that some of them are as follows: data encryption, hashing-base with hash verification, chunk signing, merkle tree chaining, traffic encryption and etc [1] [3] [5] [7]. Although these methods automatically detect fake content but they have no control over polluter sources or peers. Thus they incur great processing and communication overheads and long startup latencies due to frequent requests and retransmissions of

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damaged chunks [5] [17]. For this reason using of reputation-base security mechanisms is one of the basic requirements.

Nowadays, because of high functionality and performance trust for reputation-based systems, detection of polluter peers step is done by these systems. Many methods have been proposed in this area, but they have several strength and weakness points. Some of these methods include: black list (1), StrRepS (2), SimplyRep (3), Drank (4), Trust Management (5) and etc [2] [4] [5] [8]. The main problem of method (1) is that it is centralized, vulnerable to various attacks and has high computational overhead. Method (2) has similar disadvantages of method (1), but it is distributed. Although, the method (3) is scalable but with increasing the attacks and dynamic changing of local minimum threshold reputation, honest peers are removed from the neighbor list and reconnecting to them increases the network overhead. The biggest problem of method (4) is its Vulnerability to collusion attacks, because it uses global reputation. Table I shows comparison of proposed methods from various security aspects. The parameters used in this table are as follows: V (Vulnerable), S (Safe), N (No), Y (Yes), -- (according to the conditions is V or S).

Dealing with polluter peers is the last step in fight pollution attacks. In most reputation-base methods, if a peer reputation is less than minimum threshold reputation, then the peer is removed completely from the neighbor list. But in some methods like SimplyRep, none of peers are not removed from neighbor list to infer dynamically change threshold.

TABLE I COMPARISON PROPOSED METHODS

| Attribute/Approach | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| Whitewashing attack | V | V | S | S | S |
| Collusion attack | V | V | S | V | V |
| On-Off attack | -- | -- | S | -- | S |
| Badmouthing attack | -- | -- | -- | -- | S |
| Persistent attack | V | V | S | -- | S |
| Give chance again to polluter | N | N | Y | N | N |
| Dynamic change threshold | N | N | Y | N | N |
| Using of global reputation | Y | Y | Y | Y | N |
| Using of local reputation | Y | Y | Y | Y | N |
| Using of direct trust | N | N | N | Y | Y |
| Using of indirect trust | N | N | N | N | Y |
| Distributed approach | N | Y | Y | Y | Y |
| Centralized approach | Y | N | N | N | N |

Fig. 1 has been shown the creation and distribution of polluted content. In this figure, polluter peers create a legitimate buffer map and then send the buffer map to their neighbors on the basis of information received from neighbors and tracker server. The neighbors also, send their request to polluter peer to receive buffer map. In this step, polluter sends fake chunks to its neighbors rather than normal video chunks. Of course, this attack can be done in more complex manner that means instead of sending truly buffer map, altered buffer map and poisoning is sent by polluters. In Fig. 1, blue color indicates legitimate chunk and red color indicates polluted chunk.

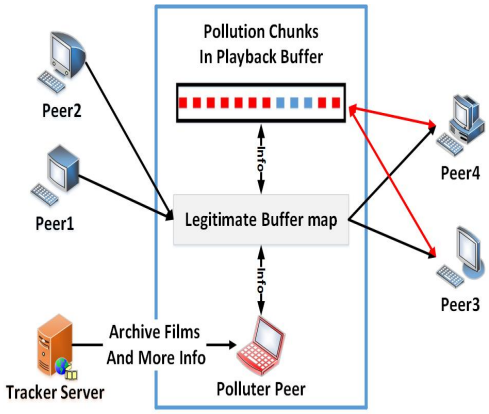


Fig. 1 The process of create and distribution polluted content in side polluter peers

## III. THE PROPOSED MECHANISM

In this paper, a novel Multi-Level mechanism is proposed to fight with pollution attacks at each of its three main steps. This mechanism mainly focuses on the detection of polluter peers step. Because, if steps 2 and 3 are performed correctly, then polluters are identified and isolated quickly and ultimately less fake content are entered into the network and peers playback buffer. The methods will be described in more detail in the following. Fig. 2 has shown general architecture that is used in proposed mechanism. In this architecture, the peers communicate with tracker servers and content distribution servers for management of their neighbours and receiving the video chunks (or frames). Peers assign to clusters based on their playback point. Multi-Level mechanism is used in this architecture for fight pollution attacks per cluster.

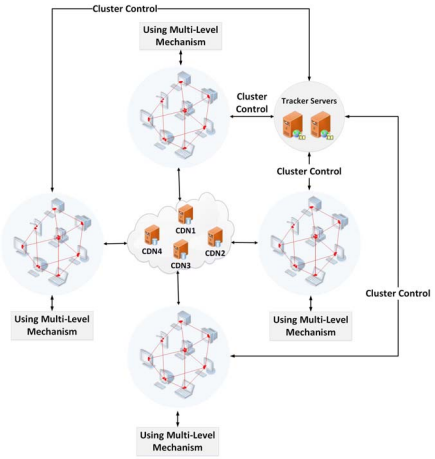


Fig.2 The general architecture of the proposed mechanism

### A. Detection Polluted Content

To detect the polluted content hashing methods are used. In this way, after network peers receive video chunks, hash value is calculated by hash functions and then the result is compared with the received hash value from the content distribution network (CDN) servers to determine whether a chunk is polluted or not.

### B. Detection Polluter Peers

For detection of polluter peers, we use reputation-based Multi-Level1, Multi-Level2 and Multi-Level3 methods. In these methods, peers compare the input chunks that neighbors do not have enough reputation with hash functions. After that neighbor peers reaches high reputation, comparison operation is done only for specific time intervals. So the operation time for comparing to hash values can be saved and this time can be used to service their neighbors. If peers take only hashing methods to fight with pollution attacks, then at the video buffering time, there is no control over the network peers and may perform more request from polluters. In addition to entering polluted chunks to peers playback buffer, there are also polluter peers in the network that reputation efficient mechanism must to be used to detect and remove of them. With the reputation mechanism, first, the request of high reputation peers is done and by this action less forged content will be in playback buffer and network.

### C. Dealing With Polluter Peers

In dealing with polluter peers, we use penalty and reward approach. In this approach, polluters are blocked in two way of temporary and permanent. Penalty process is done in specific time periods, for example if a peer is in the level *LN(n)* and fake content is distributed to neighbors, as soon as its reputation is less than of level *LLMTR(n)*, then polluter peer temporary is blocked on the *LN(n)* (n is level number in Multi-Level mechanism). Also if peers reputation is between *GMTR* to *LMTR*, polluters are given again a chance to change their behavior, otherwise if their reputation is less than *GMTR*, then they are permanently removed from the neighbor list.

### D. Multi-Level Mechanism

In this section we propose Multi-Level mechanism for fighting with pollution attack. Multi-Level is divided to three sub method called Multi-Level1, Multi-Level2 and Multi-Level3. In the proposed mechanism, four new features called *NPS*, *LN*, *PLN* and *LLMTR* are used.

Fig. 3 shows the Multi-Level mechanism and shows that each of the peers use from this mechanism independently to calculate the reputation and prioritize the neighbors of a peer are at different levels based on their reputation. The level and reputation of neighbors will change over time according to their behavior (All peers of *P1* to *Pt5* that are in levels of the Multi-Level mechanism are neighbors of a peer like *P*). First, in this mechanism, the interval *0* to *1* is discrete and each of discrete intervals is set as a level. So that, if a peer is in the level *m* ($m<n$), then created damaged content is reduced to level *m-1* and vice versa forwarded legitimate content to its neighbors rises to level *m+1*. Of course, this point should be noted that Multi-Level is used to block policy temporary in each level. For example in Fig. 3, if *Pt5* neighbor distributes fake content to *P* peer, then *Pt5* has a penalty and if its reputation is less than minimum level threshold reputation in the level *n* (or *LLMTR(n)*) it is temporarily blocked in the Level *n* until do not check for the peers remained in the level *n* and the peers in lower levels (level *n-1* to level *1*), are not requested from *Pt5* peer (In fact, as long as other peers do not investigate, do not turn to *Pt5* peer). Because, the temporary block policy prevented the further distribution of polluted content in the network. The number of levels that each peer has in Multi-Level mechanism is depends on its number of neighbors. For example, if a peer has *N* neighbors, then its number of levels is *N* and interval of each level is *1/N*. Of course, it should be noted that each peer can have variable number of neighbors, but we've considered maximum 6 neighbors for each peer.

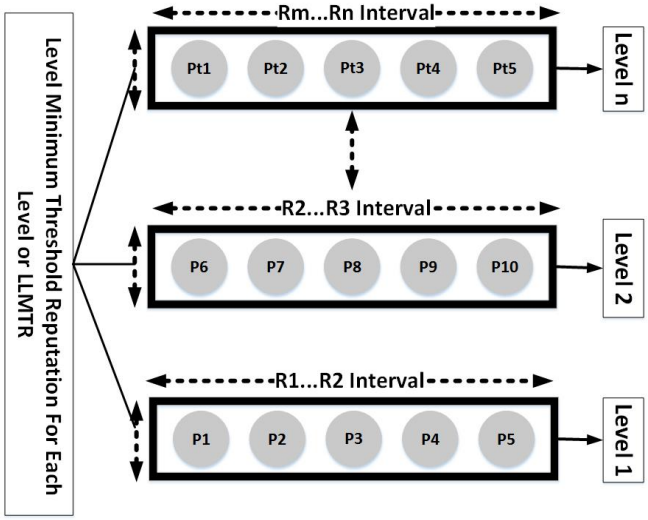


Fig.3.The Multi-Level mechanism structure in the peer as P

Peers reputation creates polluted content in Multi-Level mechanism, is calculated by Eq (1). In Eq (1) if *PEF* is greater than equal to *LMTR*, then peer is considered it a temporary honest peer otherwise, it is known as a temporary polluter peer. When a peer is the actual polluter that its reputation is less than *GMTR* and if the peer reputation is equal to *1*, then it is real honest. Peers select *Max* and *Min* by function *L* according to the pollution status of its neighbors (or *NPS*) in one of the modes, so that if *NPS* is greater than polluted threshold then they select *Min* mode and otherwise choose *Max*. Because the changing of *Min* and *Max* modes cause less forged content is distributed in network and polluter peers isolation process will be accelerated. Peers reputation that creates legitimate content, is calculated by Eq (2) (3) (4). We've used the natural world behavior in the calculation of peers' reputation. Therefore, honest peers' reputation increases slower and malicious peers' reputation reduces faster.

$$PEF=PR*((L(Min(PF,PR),Max(PF,PR)))/LN)-(PC/SC) \quad (1)$$

$$PEF1=(RF/LN)+PR \quad (2)$$

$$PEF2=(TC/SC)/LN \quad (3)$$

$$PRF=L(Min(PEF1,PEF2),Max(PEF1,PEF2)) \quad (4)$$

After each time calculation of *PEF*, peers reputation update is done to evaluate decision based on it. Update procedure is done with Eq (5).

$$PR=PEF \quad (5)$$

*1) Multi-Level1 Method*

In this method, the first neighbors list is received from tracker server and the initial reputation for all neighbors will be adjusted by the amount *LMTR*. Then, the peers get video chunks from their neighbors and a neighbor is chosen randomly based on its buffer map. Multi-Level1 method uses only local reputation to decide on the status of neighbor peers.

*2) Multi-Level2 Method*

In the second method unlike the Multi-Level1 method, peers use simultaneously the reputation and buffer map of its neighbors and then select a neighbor for service based on these two parameters.

*3) Multi-Level3 Method*

In Multi-Level3 method, when a peer comes to network, it receives a set of descriptive information about its neighbors from tracker server. Tracker calculates descriptive information based on information received from the peers in the network and provides information to help newcomer peers to find a common view of their neighbors. Descriptive information includes global reputation, the number of penalties etc (of course, it should be noted that the calculation of descriptive information can be performed by the content distribution servers). After receiving descriptive information of their neighbors ,new peers, prioritize them according to their global reputation and put in the hierarchical Multi-Level mechanism and the first requests comes from a neighbor at the highest level (highest reputation). In the Multi-Level3, global reputation descriptive is used only for new peers after that decision for detection of polluter neighbor is done based on local reputation. In this method the peers, before leaving network, sent local reputation for all of their neighbors to the tracker server and tracker based the received global reputation information of all peers calculates Eq (6) to provide descriptive information.

$$GR[i]=\sum LR[i]/PL[i] \quad (6)$$

Table II Show the parameters that been used in proposed mechanism.

TABLE II PARAMETERS OF USING PROPOSED METHODS

| | |
|---|---|
| LN | Level Number |
| LR | Local Reputation |
| GR | Global Reputation |
| PL | Partners List |
| LMTR | Local Minimum Threshold Reputation |
| GMTR | Global Minimum Threshold Reputation |
| LLMTR | Level Minimum Threshold Reputation |
| TC | True Chunks |
| PC | Polluted chunks |
| SC | Sum Chunks |
| PF | Penalty Factor |
| RF | Reward Factor |
| PEF | Peer Evaluation Factor |
| PR | Peer Reputation |
| NPUD | Number Peer Up-Down |
| NPS | Neighbors Pollution Status |

## IV. PERFORMANCE EVALUATION

In this section, we explain the types of pollution attacks performed in reputation-base systems. In the next section, we show the performance evaluation of Multi-Level mechanism against pollution attacks.

### A. On–Off Attack

On–off attack [2, 12, 13] refers to the scenario that a malicious peer sends clean and polluted chunks alternatively to the peers that request data from it. By doing this, the malicious peer can keep its reputation or trust value above the predetermined threshold, and thus avoid being identified as a polluter. The on–off attack exploits the fact that most of the trust management mechanisms are designed to tolerate certain levels of unintentionally polluted chunks (such as incomplete data chunks and erroneous data chunks) due to bad network conditions.

We used *NPUD* parameter in Multi-Level mechanism for fight with on-off attack. If changes of a peer reputation (between *LMTR* to *1*) are more than *NPUD*, then this peer is identified as a polluter and is placed in the queue according to the range of *GMTR* to *LMTR*.

### B. Collusion Attack

In this type of attack, a peer or group of malicious peers act against honest peers in the network and falsely declare negative comments about them to confirm their reputation is reduced in network. This action causes that a group of peers are honest and service or offer truly encountered with the problems and by continuing collusion attack eventually network efficiency reduces and honest peer must be leave the network [6].

By using the temporary blocked policy in Multi-Level, polluter peers with high reputation are blocked in top level and honest peers have opportunity that with lower reputation can reach highest reputation levels.

### C. Whitewashing Attack

Because of distribution of fake data in network, the reputation of malicious peers is less than minimum threshold reputation and they must leave the network (any defense method based on reputation, considers itself a minimum threshold reputation and due to its decision about pollution of a peer). But these malicious peers entered the network again at another time with new identity (ID) after leaving the network and initiate their reputation to the value that is more than the minimum threshold reputation. So, they can again send content to network but because these peers are polluter instead of sending legitimate content, send forged content.

Due to the use of temporary blocked policy, as soon as a polluter peer obtains a new ID and send fake content in the network, immediately is blocked at its level and most likely is placed in the range of *GMTR* to *LMTR*. So, since in our mechanism newer peers reputation is greater than equal to *LMTR,* defense with whitewashing attacks is performed easily in Multi-Level mechanism.

### D. Simulation Framework

To perform evaluation of the Multi-Level mechanism through network simulation, the OMNeT++ v.4.1 [10] and INET framework [9] are used. The INET framework is used for simulating TCP/IP networks in OMNET++. INET framework implements UDP, IP and Data Link Layer in OMNeT++. OverSim [18] is a framework in OMNeT++ for simulating P2P systems. OverSim uses INET framework to simulate underlay layers. For this simulation, we used the CDN-P2P structure for streaming VoD that is proposed in [16]. This system uses central tracker to create a cluster mesh structure. Each cluster has two direct links to the CDN server. Peers arrival process is Poisson distribution with mean inter arrival time of 2 seconds. Each peer selects upload bandwidth with uniform distribution between 350 Kbps to 2 Mbps and downloads bandwidth between 1 to 4 Mbps.In this simulation the Verbose_ARDTalk sample video trace file [11] is used.

Table III shows the rest of parameters that is used for simulation of proposed mechanism. Parameter values were obtained based on the simulation.

TABLE III SIMULATION PARAMETERS

| Parameter | Value |
|---|---|
| NPS | 2 |
| LMTR | 0.2 |
| GMTR | 0.05 |
| Pollution Percent | 1-10 % |
| PF | 0.019-0.01938 |
| RF | 0.016-0.01644 |
| NPUD | 2 |
| Number of peers | 200 |
| Maximum packet size | 1000 Bytes |
| Video codec | MPEG4 Part I |
| Video FPS | 25 |
| Number of frames in GoP | 12 frames |
| Length of films | 30 minutes |
| Startup buffering | 12 s (300 frame) |
| Selected trace file | Verbose_ARDTalk |
| Average video bit rate | 512 Kbps |
| Chunk size | 1 frame |
| Number of CDN servers | 4 |
| Mean neighbor size of each peer | 6 |

### E. Overhead Rate

Fig. 4 to 6 show overhead of Multi-Level mechanism for various attack scenarios. Also, this figure shows Multi-level method detects polluter peers with less overhead compared to SimplyRep and DRank.

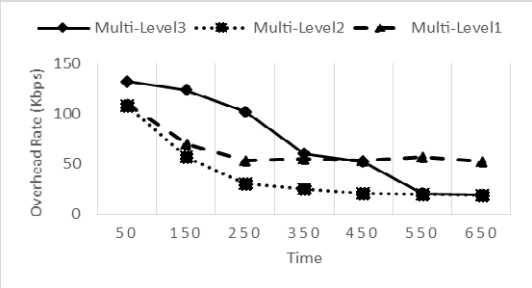


Fig. 4 Average overhead of pollution attacks detection methods with collusion attack (% 10 of polluters)

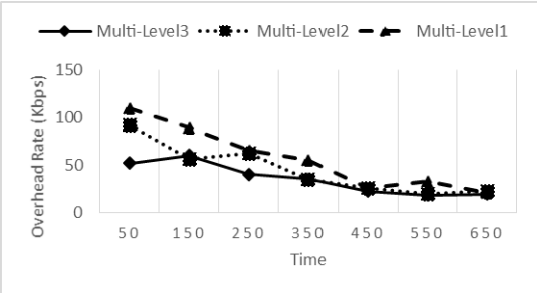


Fig. 5 Average overhead of pollution attacks detection methods with on-off attack (% 10 of polluters)

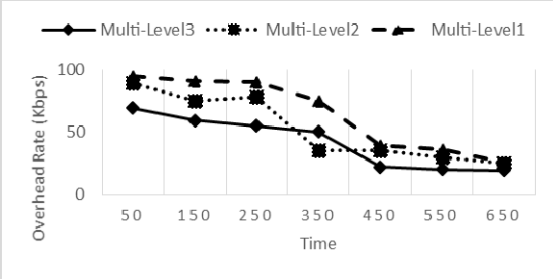


Fig. 6 Average overhead of pollution attacks detection methods with whitewashing attack (% 10 of polluters)

### F. Detection Rate

Fig.7 to 9 show the performance of Multi-Level against collusion attack, whitewashing attack and on-off attack. Generally, for polluter detection rate of various attacks Multi-Level3 has better performance than Multi-Level1 and Multi-Level2.

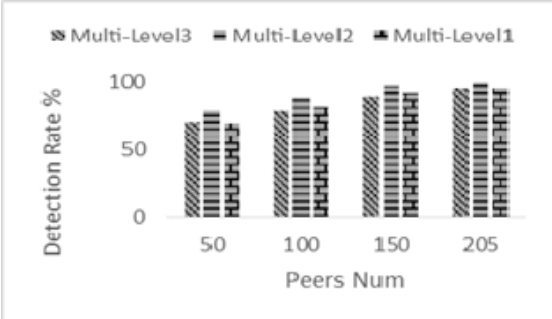


Fig. 7 Detection rate of polluter peers with collusion attack (% 10 of polluters)

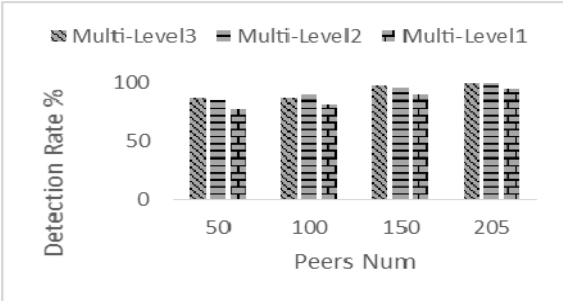


Fig. 8 Detection rate of polluter peers with whitewashing attack (% 10 of polluters)

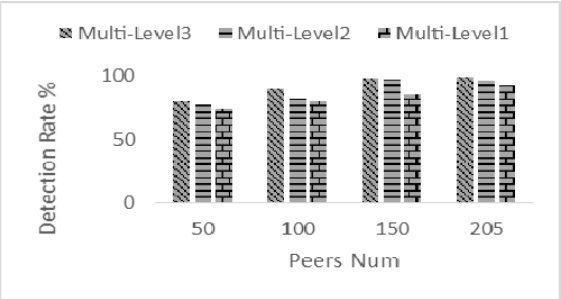


Fig. 9 Detection rate of polluter peers with on-off attack (% 10 of polluters)

## V. CONCLUSION AND FUTURE WORKS

Nowadays, companies use live and VoD streaming services on P2P architecture for using clients' resource and reduction of their costs. But, many attacks to threat the architecture are launched by malicious peers in the network. One of these attacks that is happened in video streaming, is pollution attack. In this type of attack, polluters alter video chunks to fake chunks and then reduce the performance of system by distributing of forged chunks in network. In this article, we proposed the Multi-Level mechanism as a distributed reputation-base system to fight pollution attacks in P2P streaming. This mechanism is divided to three methods: Multi-Level1, Multi-Level2 and Multi-Level3. Multi-level mechanism is dependent on the hash-based techniques to detect polluted content, but these techniques cannot detect and dealing with polluters (steps 2 and 3 of pollution attacks) alone due to increasing of chunks retransmission overhead. So, Multi-Level is used with temporary blocked policy and new reputation calculation techniques in distributed manner to fight pollution attacks. The results showed that our mechanism efficiently fights with collusion, whitewashing and on-off attacks and has desirable performance.

For future works, we want to examine other methods to calculate peers reputation. In addition, we will use a method named to "Hash-Distribution By Peers". With this method, better detection of polluted content and less come to servers overhead is achieved.

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